

Python Math Operations

A simple calculator script demonstrating addition, subtraction, multiplication, and division.

1. Overview

The program below defines four functions — **add**, **subtract**, **multiply**, and **divide** — each taking two numbers as input. It also includes a **divide-by-zero** check and a simple command-line interface that prompts the user for two numbers and prints the results of all four operations.

2. Source Code (calculator.py)

```
"""
Simple Calculator
Performs addition, subtraction, multiplication, and division on two numbers.
"""

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Error: Division by zero is not allowed"
    return a / b

def main():
    print("=== Simple Calculator ===")

    try:
        num1 = float(input("Enter the first number: "))
        num2 = float(input("Enter the second number: "))
    except ValueError:
        print("Invalid input. Please enter numeric values.")
        return

    print(f"\nAddition:      {num1} + {num2} = {add(num1, num2)}")
    print(f"Subtraction:    {num1} - {num2} = {subtract(num1, num2)}")
    print(f"Multiplication: {num1} * {num2} = {multiply(num1, num2)}")
    print(f"Division:       {num1} / {num2} = {divide(num1, num2)}")

if __name__ == "__main__":
    main()
```

3. Sample Run

```
=== Simple Calculator ===  
Enter the first number: 10  
Enter the second number: 5  
  
Addition:      10.0 + 5.0 = 15.0  
Subtraction:   10.0 - 5.0 = 5.0  
Multiplication: 10.0 * 5.0 = 50.0  
Division:      10.0 / 5.0 = 2.0
```

4. How to Run

Save the code as **calculator.py** and run it from a terminal using the command: `python3 calculator.py`. Enter two numbers when prompted to see the results of all four operations.